التاريخ: / / الموضوع:

out h_1 = .24834 t_1 = .01 η = 5
out h_2 = .24949 t_2 = .99
out o_1 = .8062
out o_2 = .8379

$$\frac{\partial E}{\partial w_5} = \frac{\partial E}{\partial out_{o_1}} \cdot \frac{\partial out_{o_1}}{\partial net_{o_1}} \cdot \frac{\partial net_{o_1}}{\partial w_5}$$

$\delta o_1 = (out_{o_1} - t_1)(1 - out^2_{o_1}) =$
$(.8062 - .01)(1 - .8062)^2 = .2787$

$w_{new} = w_{old} - \eta\,(\delta * output)$

$w5 = .40 - 0.0597 \simeq .3327$

$= (out_1 - t_1)(1 - out^2_{o_1})(out\,h_2)$
$.7962 * .35004156 * .24949 =$
$.1379$

$6 = .45 - .5(0.1379) = .3811$

التاريخ: / /　　　　　　　　　　　　الموضوع: ____________

Assignment 2

Back propagation

$\eta = 0.5$ → learning rate　　　　　　Assumed doctor lee ←

out h_1 = .4834　　　　t_1 = .01

out h_2 = .4949　　　　t_2 = .99

out o_1 = .8062

out o_2 = .8379

$$\frac{d}{dx}(\tanh(x)) = 1 - \tanh^2(x)$$

$$\partial o_1 = (\text{out}_{o_1} - t_1)(1 - \text{out}^2_{o_1})$$

$$= (.8062 - .01)(1 - 0.8062^2)$$

$$.7962 \times 0.35004156 = ~~[illegible]~~ \; .2787$$

$$\partial o_2 = (.8379 - .99)(1 - .8379^2) =$$

$$(-0.1521)(0.29792359) \simeq$$

$$-0.0453$$

$$w_{new} = w_{old} - \eta(\partial \times \text{output})$$

$$w_5 \text{ new} = .40 - 0.5(~~[illegible]~~ \; .2787 \times .4834) =$$

$$.40 - 0.0597 \simeq ~~[illegible]~~ \; .3327$$

$$w_6 \text{ new} = .45 - .5(~~[illegible]~~ \; .2787 \times 0.4949)$$

~~[illegible]~~

$$.45 - 0.0611 \simeq ~~[illegible]~~ \; .3811$$

$w_7\ new = .50 - .5(-0.0453 * .4834)$

$.50 - (-0.01095) \simeq 0.5110$

$w_8\ new = .55 - .5(-0.0453 * .4949) =$

$w_8\ new = .55 - (-0.0112) \simeq 0.5612$

$\partial h_1 = (1 - out^2 h_1)[(\partial_{o_1} * w_5) + \partial_{o_2} * w_7)]$

$= (1 - 0.4834^2)[(.2787 * .40) + (-0.0453 * .5)]$

$= .76632444 * 0.08883 \simeq 0.0681$

$\partial h_2 = (1 - out^2 h_2)[(\partial_{o_1} * w_6) + (\partial_{o_2} * w_8)]$

$= (1 - .4949^2)[(.2787 * .45) + (-0.0453 * .55)]$

$= .75507399 * .1005 \simeq .0759$

$w_1\ new = .15 - .5(0.0681 * .05) =$

$.15 - .0017025 \simeq .1483$

$w_2\ new = .20 - .5(.0681 * .10) =$

$.20 - .00292 \simeq .1966$

$w_3\ new = .25 - .5(.0759 * .05) \simeq$

$.2481$

$w_4\ new = .30 - .5(.0759 * .10) =$

$.30 - .003795 \simeq .2962$